

Exploratory Data Analysis Report

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Basics

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Recorder: Barrett Viator (Week 4)

Title: *Will Bridging The Digital Divide Bring Prosperity To The Second “Next Billion Users” And Their Home Countries?*

GitHub: <https://github.com/Merrimack-Capstone/MSDS-capstone>

Recap: Background & Question

- **We are posing the following research question:** *To what extent has growth in internet access of populations influenced population wellbeing in more developed nations, and can that be used to predict improvements in wellbeing and economic progress among global minority populations (the "next billion users")?*
- **Our hypothesis is as follows:** *Increased access to the internet in the developing world is a measurably material contributing factor to improved population wellbeing - so much so that we may be able to predict how much wellbeing will improve based on increased digital inclusion.*
- **Our predictions:** *We expect that digital inclusion will be strongly correlated to education, which is a component of wellbeing, and moderately correlated with overall wellbeing. We expect that PREDICTING the rate of improvement in wellbeing based on internet inclusion*

alone will be extremely challenging due to the number of other factors influencing wellbeing, but hope to be able to isolate its impact on wellbeing. We expect that clustering may help lead to better predictive performance, if we have time to get to it.

Methods used for Exploratory Analysis:

- **Datasets and acquisition methods**

- We are using the following four data sets, with automated data acquisition as noted (see Appendix A for a full data dictionary):
 - World Bank: Global Development Indicators
 - *Contains:* A subset (curated by us) of development indicators by country and year, going back to 1995, for well over 200 countries, which contains data we believe will impact wellbeing (features), data we could use to define a wellbeing index should we choose to do so (responses) and income group categorization which we believe may be useful controls and in clustering.
 - *Acquisition Method:*
 - Features and responses: API calls to directly retrieve data from the World Bank work via the Python library “wbgapi” where columns are retrieved via a series code that corresponds to varying topics covered by data in the database. These series codes can be further queried to be filtered by the range of years and economies that are relevant to an individual's use cases, but they must also be renamed more descriptively, as the indicator codes for the API do not give much insight on their own.

- Income groupings: Spreadsheet download from World Bank, extracted and saved as a CSV (for reasons outlined in footnote below)¹
- *Data Format:* Imported directly into a dataframe in R
- UNDB: Wellbeing and income inequality indicators
 - *Contains:* The HDI index of wellbeing, the Gini index of income inequality, and the IHDI inequality- adjusted wellbeing index
 - *Acquisition Method:* The UNDP database requires an API key that can be easily obtained for free via a registration form. This API process is different from the one seen by the World Bank and has much more strict limitations on how much data can be retrieved by one call. In order to alleviate this constraint, we produced fifteen queries to retrieve the information that we needed for the range of years being explored in our study. Many of the fields returned N/A, as some of this data was not collected historically as far back as we queried or only produced in specific years.
 - *Data Format:* Imported directly into a dataframe in R
- ITU: Mobile and broadband data (optional and not part of EDA - will be dealt with towards end of project if time permits)
 - *Contains:* Mobile and broadband internet access data by country / year
 - *Acquisition Method:* Data from the ITU is acquired through links on their website that directly download the csv files. In order to retrieve these files directly into the code, the download link must be retrieved from the developer explorer in a

¹ The Income Groupings in the World Bank data are not a direct series name that can be extracted via API call, but rather are derived by the World Bank using rules applied to data in their database. In the interest of time, we chose to download a spreadsheet from their site that had the groupings by country on it and save that data as a CSV. Given more time, we would replicate that grouping derivation or create our own.

browser and the Python library IO in order to take the downloaded file and pass it into a dataframe. There is no other direct API process to retrieve this information as seen with World Bank and UNDP.

- *Data Format:* Imported directly into a dataframe in R
- Since our EDA is being performed in R, but the API calls need to be written in Python, the data acquisition was automated using the “reticulate” library of functions which permits running Python code from within R. This allowed us to run the API calls and import the data directly into R data frames, rather than have to run Python code, pull the data, then save it to static csv files to be imported into R, which is inefficient and inelegant.
- The World Bank and UNDB data were joined into one data frame within the reticulate code using the Python pandas merge function, with a left join between the ISO3 country code and year as common fields. In order to produce an ISO3 code that could be accessed for this join, the economy column had to be split because it included both an ISO3 identifier and a country name. The pycountry library was used to then map the country name into the combined dataframe to create a country column for ease of reference later. Since ISO3 country codes are standardized this was fairly straightforward.
- The income level data (from the spreadsheet we saved) was joined into the data frame within R using the left_join function, using Country Code as the shared field
- Since the API calls take 15-20 minutes to complete, we separated this code from the EDA code, ran it and stored the resulting dataframe in an R “.rds” file, which is then imported into our EDA file. This is only for convenience while we develop our EDA code so we don’t need to constantly re-run the API data calls (in case our session is terminated and the

data is no longer in our R environment). For production code, the data will be directly loaded via API.

- **Exploratory Analysis Methods:**

We generated a range of tables and visualizations to explore the data and inform feature selection, engineering, dimensionality reduction, and model design. These outputs, detailed in the “Tables, Visualizations, and Brief Interpretations” section, cover the following categories:

- *Descriptive Statistics:* Basic tabular analysis of every variable, primarily using `summary()` and `skim()`, was used to understand data distribution, range and missing values
- *Data Visualizations:* We used histograms, violin plots, scatterplots, bar charts, and heatmaps (through `ggplot2`) to visualize relationships and correlations between variables, data distribution / skewness challenges, potential clustering / control variables, and data sparseness / missingness, and to perform an initial basic test of whether our hypothesis has a chance of being proven.
- *Correlation Analysis:* We used visualized (heatmap) and numeric correlation analysis tools, both filtered and unfiltered, to explore the correlations between features, both to identify features that are likely to be predictive and to identify potential collinearity, which can guide our decisions on features to drop and/or combine to reduce dimensionality and reduce the impact of collinearity on statistical model variance.
- *Missing Data Exploration:* We calculated the proportion of missing data across two axes (by variable, and by year) and visualized the results. These visualizations help identify patterns in missing data to guide decisions on how to fill in (e.g. interpolate missing years). They also help guide decisions on which variables and/or entire year sets may simply be too sparse and should be dropped entirely/

- *Time Series Diagnostics:* To incorporate the time dimension that was absent from earlier analyses, we computed annual averages of Internet Users (%), HDI, and IHDI, both globally and by income group. This allowed us to detect trends and turning points over time that are not visible when pooling all years together.
- **Use of Generative AI:** As indicated in our Final Proposal, we used Generative AI to assist with producing the EDA, in two ways:
 - *Primary Use:* We used it to refine and supplement our EDA strategy (See Appendix B). While many recommendations aligned with our initial plans, it provided valuable guidance on visualizing correlation and sparseness, which we had previously addressed only in tabular form.
 - *Secondary Use:* We used it to assist with debugging and refining ggplot-related code (solely under human direction and validation). This enabled rapid, accurate parameterization of complex visualizations by streamlining syntax, resolving errors, and accelerating iteration far more efficiently than researching reference tools like GeeksForGeeks and Stack Overflow. This proved especially useful given ggplot's steep learning curve and its extensive, often finicky, parameterization.

Discussion: We find that the visualization of the data, found in Figures 1-10 is much more useful than the tabular data in revealing insights. Those insights include:

- Our initial exploration of the relationship between digital inclusion and wellbeing, shown by the upward trendlines in Figures 1 and 2, indicates that the hypothesis has merit. However, the consistently low R^2 values across income clusters highlight a key

limitation: internet access alone is not a strong predictor of overall wellbeing. This finding aligns with our prediction that numerous other socio-economic, health, and governance factors contribute significantly to a country's wellbeing, particularly in lower-income nations where the predictive power of internet access is even more diminished despite a steeper correlation. This complexity highlights the necessity of considering a multi-faceted approach to understanding and modeling wellbeing beyond a singular focus on digital connectivity, and may also lead to alternative ways to represent and model this singular relationship as well.

- Further in-depth analysis of the data reveals significant distributional challenges, particularly a pronounced left skew with a large cluster of data points at or near 0% internet access, as can be seen in figures 3 and 4. This issue is more prevalent in low and middle-income countries, likely reflecting their later adoption of internet technologies. This clustering poses a challenge for predictive modeling, as the left-skewed distribution may introduce significant variance that distorts true relationships. Addressing this will likely require data transformation or specialized modeling techniques.
- Differences in data distribution by income group should be considered when refining our models. Figure 5 demonstrates marked differences (and positive correlation) between income group and wellbeing. This is expected, given that Gross National Income per capita is a core component of the HDI index—though it does raise questions about the chart's standalone utility. However, taken in concert with Figure 6, which shows the stark differences in median level and distribution of internet access by income group, we can infer a logical progression that is central to our thesis which is that internet access is highly correlated to income which in turn contributes significantly to wellbeing. So

these exhibits not only further support our hypothesis but suggest that segmenting our data by income group may lead to better model results and prescriptive recommendations for the lower income cohorts.

- The identification of both strong positive and negative correlations, as represented in Figures 7-9, is essential for addressing potential multicollinearity in multivariate models. These highly correlated variables warrant further investigation because they can lead to unstable coefficient estimates and inflated standard errors. This demonstrates the need for informed feature selection or dimensionality reduction techniques to build more accurate and interpretable models of wellbeing. Figures 8 and 9 help identify which features should be removed entirely versus those that warrant further evaluation using VIF analysis.
- A comprehensive assessment of data sparseness (“missingness”) provides essential insights for subsequent steps. We can see in Figures 10-12 that some variables individually, and year series in their entirety, exhibit such extensive missingness that should likely be dropped. It’s also evident that there’s a general trend of improved data completeness over time, and that there are certain patterns in missingness that will guide our strategy for dealing with null values (e.g. by interpolating values when data is only available every other year). One additional, particularly important finding, is that the Gini index and inequality-adjusted IHDI index don’t appear in the data until 2010 - so we may need to reconsider whether using this as our response to factor in the effect of income inequality is still appropriate given the smaller data set available for this.²

² This concern is in addition to broader concerns raised in peer review that “handicapping” wellbeing using income inequality introduces another complicating factor to our approach that may confound our modeling

- Finally, the time series analyses reveal *absolutely crucial* insights which at a minimum reinforce that internet access alone cannot be used to predict wellbeing, and may even suggest that our thesis is flawed, at least for the very low income countries which motivate this study. Figure 13 shows global trends over time in average internet usage % and HDI (scaled separately for visual clarity). The upward trajectory of both plots (save for HDI which briefly fell during the pandemic years of 2020-2021 before recovering) suggests a strong correlation between the change in internet usage and wellbeing. HOWEVER - if we break this out into Income Group cohorts as shown in Figure 14, it would appear that, for Low Income countries, HDI grew steadily until 2009 despite little to no internet access, whereas from 2010 on, HDI growth slowed visibly, just as internet access really begins to ramp up. Indeed, if we zero in on this cohort as shown in Figure 15, we see this quite clearly. We can conjecture on the reasons for this: First, during the period beginning in 2009, the world fell into the Great Recession, which disproportionately impacted lower income countries both in its severity and duration - these additional impacts may simply be offsetting much of the benefit of expansion of internet access during this period. In addition - as Figure 15 shows, the impact of inequality in these Low Income countries was severe during this time period - the IHDI index (already much lower than the HDI index suggesting that inequality is severe in these countries) actually declines and doesn't recover for ten years. This may simply suggest that the benefits of improved internet access, along with other economic benefits, are simply accruing to a too-narrow subset of the population.

Next Steps:

Our initial analysis focused on intensive EDA to evaluate the quality and suitability of the datasets. Moving forward, we will proceed with the handling of missing values, reducing dimensionality via PCA, and addressing multicollinearity through VIF. Outlier detection and treatment are also crucial to the success of our modeling efforts. Utilizing the interquartile range (IQR) to identify and cap extreme values in key numerical columns will mitigate their disproportionate influence on model performance. Data types will need to be assessed to convert to appropriate formats for modeling which will ensure consistent handling of numerical and categorical data throughout the following modeling stages.

Feature engineering will then be systematically applied to enhance the predictive power of our dataset by transforming existing variables and creating new binned ones to provide deeper insights into the various related columns and to reduce noise that may impact model performance. Planned feature engineering steps include:

- Defining an alternative response variable that converts the numeric HDI feature to binned states of wellbeing - given the anticipated low R^2 , a categorical model that predicts shifts higher wellbeing states based on digital inclusion may be more accurate and lead to better prescriptive recommendations than a model that attempts to predict HDI outright
- Redefining our key variables (internet access and HDI) from their absolute levels to their period-to-period change - this may help improve the focus on causality and shift the model to prediction of **contribution** of growth in internet access to growth in HDI rather than in a straight 1:1 predictive relationship. It may also help address the extreme left skew (near-zero internet access) and support incorporation of time-lag effects into future models.

- Defining additional variables that factor in time-lag; Since investment in internet access today may take time to influence positive wellbeing, defining additional response variables that calculate the change in HDI that occurs N years from now may be a useful modeling technique.

In order to prepare the features for machine learning models, scaling and encoding techniques will be applied. Numerical features often appear with varying scales will undergo standardization using StandardScaler to ensure that no single feature dominates the machine learning process. While all our variables (except Income Level, which is a control rather than a feature or response) are currently numeric, one-hot encoding will be utilized for any nominal categorical variables we create that do not exhibit an inherent order by creating columns for each unique category and preventing unintended ordinal relationships.

This comprehensive pre-processing and feature engineering pipeline ensures that the data is properly cleaned and well-structured for predictive modeling and other machine learning algorithms. This will lay a stronger foundation for the subsequent model training and evaluation phases to test our hypothesis and to compare the relationship of the results to our predictions. It also sets the stage for clustering and building different models for different country segments, as suggested by the anomalies revealed in the time series analysis.

Tables, Visualizations, and Interpretations

Table 1: Summary Statistics (Variables 1 to 8)

CountryCode	CountryName	year	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore
Length:5974	Length:5974	Min. :1995	Min. :0.2380	Min. :0.1340	Min. : 4.819	Min. : 17.00	Min. :-1.96956
Class :character	Class :character	1st Qu.:2002	1st Qu.:0.5700	1st Qu.:0.3990	1st Qu.:11.060	1st Qu.: 76.58	1st Qu.: -0.82041
Mode :character	Mode :character	Median :2009	Median :0.7110	Median :0.5930	Median :19.183	Median : 94.52	Median :-0.29892
NA	NA	Mean :2009	Mean :0.6899	Mean :0.5783	Mean :21.317	Mean : 91.33	Mean :-0.05974
NA	NA	3rd Qu.:2016	3rd Qu.:0.8140	3rd Qu.:0.7500	3rd Qu.:31.878	3rd Qu.:102.80	3rd Qu.: 0.59246
NA	NA	Max. :2023	Max. :0.9720	Max. :0.9230	Max. :51.891	Max. :578.71	Max. : 2.45912
NA	NA	NA	NA's :422	NA's :3677	NA's :3670	NA's :720	NA's :1204

Table 2: Summary Statistics (Variables 9 to 16)

ElectricAccess	PopDensity	PopInSlums	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex
Min. : 0.80	Min. : 1.438	Min. : 0.00	Min. : 0.0258	Min. : 0.40	Min. :0.0054	Min. :-2.44023	Min. : 0.6032
1st Qu.: 63.20	1st Qu.: 30.875	1st Qu.: 1.55	1st Qu.: 3.7148	1st Qu.: 31.07	1st Qu.:0.2383	1st Qu.: -0.78651	1st Qu.: 7.5392
Median : 98.10	Median : 77.922	Median :21.18	Median : 11.1751	Median : 54.05	Median :0.6029	Median :-0.20432	Median : 15.1677
Mean : 79.23	Mean : 298.200	Mean :28.73	Mean : 65.0840	Mean : 57.04	Mean :0.9675	Mean :-0.06367	Mean : 25.1422
3rd Qu.:100.00	3rd Qu.: 176.610	3rd Qu.:52.80	3rd Qu.: 39.6974	3rd Qu.: 86.76	3rd Qu.:1.3904	3rd Qu.: 0.59433	3rd Qu.: 34.1564
Max. :100.00	Max. :18822.891	Max. :99.81	Max. :3850.5000	Max. :100.00	Max. :6.0192	Max. : 2.46966	Max. :171.1775
NA's :627	NA's :583	NA's :4011	NA's :1505	NA's :5412	NA's :3666	NA's :1223	NA's :3592

Table 3: Summary Statistics (Variables 17 to 24)

FixedIntSubs	InternetUsersPet	PoliticalStability	RuleOfLaw	RqdEduYears	GovtEduSpendPetGDP	MortalityFromDirtyness	UHCServiceCoverage
Min. : 0.0000	Min. : 0.00	Min. :-3.31295	Min. :-2.59088	Min. : 4.00	Min. : 0.000	Min. : 0.40	Min. :11.00
1st Qu.: 0.2673	1st Qu.: 2.81	1st Qu.: -0.69566	1st Qu.: -0.82174	1st Qu.: 8.00	1st Qu.: 3.139	1st Qu.: 2.95	1st Qu.:44.00
Median : 3.8683	Median : 21.00	Median : 0.02612	Median :-0.20532	Median : 9.00	Median : 4.232	Median : 6.10	Median :63.00
Mean :10.5358	Mean : 32.58	Mean :-0.06279	Mean :-0.06373	Mean : 9.42	Mean : 4.418	Mean : 16.96	Mean :59.39
3rd Qu.:18.4017	3rd Qu.: 61.40	3rd Qu.: 0.79477	3rd Qu.: 0.69307	3rd Qu.:11.00	3rd Qu.: 5.428	3rd Qu.: 25.55	3rd Qu.:75.00
Max. :62.2147	Max. :100.00	Max. : 1.75868	Max. : 2.12476	Max. :17.00	Max. :15.863	Max. :108.10	Max. :91.00
NA's :2041	NA's :675	NA's :1160	NA's :1114	NA's :1432	NA's :2288	NA's :5791	NA's :4630

Table 4: Summary Statistics (Variables 25 to 32)

HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPet	FoodInsecurityPet	RuralPopulGrowth	VoiceAccountability	Homicides	Income Group
Min. : 4.176	Min. : 0.1489	Min. : 0.036	Min. : 2.00	Min. :-235.9259	Min. :-2.31340	Min. : 0.000	Length:5974
1st Qu.: 63.631	1st Qu.: 21.9384	1st Qu.: 1.212	1st Qu.: 8.90	1st Qu.: -0.5789	1st Qu.: -0.88313	1st Qu.: 1.228	Class :character
Median : 261.295	Median : 128.2416	Median : 3.663	Median :22.30	Median : 0.4574	Median :-0.00619	Median : 2.861	Mode :character
Mean : 959.978	Mean : 648.7002	Mean : 8.791	Mean :29.09	Mean : 0.3933	Mean :-0.04756	Mean : 7.939	NA
3rd Qu.: 890.483	3rd Qu.: 557.3557	3rd Qu.:10.632	3rd Qu.:45.30	3rd Qu.: 1.6167	3rd Qu.: 0.85004	3rd Qu.: 8.879	NA
Max. :12434.434	Max. :7871.2450	Max. :88.404	Max. :89.20	Max. : 15.0306	Max. : 1.80099	Max. :138.774	NA
NA's :1617	NA's :1630	NA's :5009	NA's :4953	NA's :458	NA's :1112	NA's :2755	NA

COMMENTS:

- Basic summary of every variable, showing type, distribution, mean/median, and missingness
- Can use this to start understanding our data better and to deal with sparseness
- Pretty dense, but we can already see a few things:
 - We have 400+ observations without our core HDI_Index response
 - We'll need to reconsider whether to use IHDI given over 60% is missing
 - There are other fields with large % missing (too sparse)

Table 5: Data summary

Name	FirstCutData
Number of rows	5974
Number of columns	32
Column type frequency:	
character	3
numeric	29
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CountryCode	0	1.00	3	9	0	206	0
CountryName	319	0.95	4	38	0	195	0
Income Group	377	0.94	10	19	0	4	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	2009.00	8.37	1995.00	2002.00	2009.00	2016.00	2023.00	
HDI_Index	422	0.93	0.69	0.16	0.24	0.57	0.71	0.81	0.97	
IHDI_Index	3677	0.38	0.58	0.20	0.13	0.40	0.59	0.75	0.92	
GiniCoeff	3670	0.39	21.32	11.35	4.82	11.06	19.18	31.88	51.89	
FoodIndex	720	0.88	91.33	28.06	17.00	76.57	94.52	102.80	578.71	
CorruptionScore	1204	0.80	-0.06	1.00	-1.97	-0.82	-0.30	0.59	2.46	
ElectricAccess	627	0.90	79.23	30.10	0.80	63.20	98.10	100.00	100.00	
PopDensity	583	0.90	298.20	1389.80	1.44	30.87	77.92	176.61	18822.89	
PopInSlums	4011	0.33	28.73	27.32	0.00	1.55	21.18	52.80	99.81	
WaterStress	1505	0.75	65.08	266.68	0.03	3.71	11.18	39.70	3850.50	
BankingAccess	5412	0.09	57.04	30.13	0.40	31.06	54.05	86.76	100.00	
RnDInvest	3666	0.39	0.97	0.98	0.01	0.24	0.60	1.39	6.02	
GovtEffectiveness	1223	0.80	-0.06	1.00	-2.44	-0.79	-0.20	0.59	2.47	
ShippingIndex	3592	0.40	25.14	25.22	0.60	7.54	15.17	34.16	171.18	

COMMENTS:

- Similar to the Summary data above
- Adds a “complete_rate” variable which shows what % of the data is populated, which is super helpful

Table 6: Correlation Matrix: Columns 1 to 8

	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore	ElectricAccess	PopDensity	PopInSlums
HDI_Index	1.00	0.98	-0.92	0.29	0.71	0.85	0.17	-0.82
IHDI_Index	0.98	1.00	-0.97	-0.10	0.72	0.80	0.14	-0.85
GiniCoeff	-0.92	-0.97	1.00	0.10	-0.62	-0.79	-0.09	0.84
FoodIndex	0.29	-0.10	0.10	1.00	0.13	0.20	0.08	-0.20
CorruptionScore	0.71	0.72	-0.62	0.13	1.00	0.47	0.20	-0.57
ElectricAccess	0.85	0.80	-0.79	0.20	0.47	1.00	0.10	-0.75
PopDensity	0.17	0.14	-0.09	0.08	0.20	0.10	1.00	-0.11
PopInSlums	-0.82	-0.85	0.84	-0.20	-0.57	-0.75	-0.11	1.00
WaterStress	0.12	0.08	-0.08	0.01	0.04	0.13	0.05	-0.14
BankingAccess	0.82	0.81	-0.72	0.04	0.75	0.56	0.18	-0.69
RnDInvest	0.63	0.62	-0.47	0.12	0.69	0.27	0.07	-0.47
GovtEffectiveness	0.79	0.83	-0.74	0.12	0.92	0.57	0.22	-0.64
ShippingIndex	0.46	0.46	-0.38	0.10	0.35	0.34	0.34	-0.36
FixedIntSubs	0.73	0.88	-0.80	0.14	0.64	0.47	0.24	-0.57
InternetUsersPct	0.72	0.87	-0.82	0.29	0.56	0.55	0.15	-0.64
PoliticalStability	0.59	0.63	-0.58	0.10	0.75	0.41	0.14	-0.56
RuleOfLaw	0.74	0.77	-0.68	0.13	0.94	0.50	0.18	-0.63
RqdEduYears	0.41	0.39	-0.38	0.13	0.26	0.42	-0.01	-0.36
GovtEduSpendPctGDP	0.26	0.28	-0.28	0.11	0.35	0.20	-0.15	-0.33
MortalityFromDirtness	-0.80	-0.79	0.80	0.15	-0.42	-0.82	-0.07	1.00
UHCServiceCoverage	0.92	0.92	-0.86	0.25	0.61	0.81	0.13	-0.79
HealthSpendPerCapita	0.64	0.68	-0.57	0.09	0.73	0.36	0.26	-0.48
GovtHealthSpendPerCapita	0.63	0.67	-0.57	0.09	0.72	0.35	0.29	-0.47
MigrantsPct	0.41	0.50	-0.41	0.06	0.43	0.34	0.37	-0.38
FoodInsecurityPct	-0.83	-0.85	0.84	0.14	-0.62	-0.78	-0.13	0.77
RuralPopulGrowth	-0.24	-0.52	0.50	-0.02	-0.30	-0.23	-0.03	0.48
VoiceAccountability	0.59	0.65	-0.58	0.10	0.77	0.40	0.08	-0.55
Homicides	-0.24	-0.30	0.26	-0.01	-0.24	-0.06	-0.09	0.17

Table 7: Correlation Matrix: Columns 9 to 16

	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex	FixedIntSubs	InternetUsersPct	PoliticalStability
HDI_Index	0.12	0.82	0.63	0.79	0.46	0.73	0.72	0.59
IHDI_Index	0.08	0.81	0.62	0.83	0.46	0.88	0.87	0.63
GiniCoeff	-0.08	-0.72	-0.47	-0.74	-0.38	-0.80	-0.82	-0.58
FoodIndex	0.01	0.04	0.12	0.12	0.10	0.14	0.29	0.10
CorruptionScore	0.04	0.75	0.69	0.92	0.35	0.64	0.56	0.75
ElectricAccess	0.13	0.56	0.27	0.57	0.34	0.47	0.55	0.41
PopDensity	0.05	0.18	0.07	0.22	0.34	0.24	0.15	0.14
PopInSlums	-0.14	-0.69	-0.47	-0.64	-0.36	-0.57	-0.64	-0.56
WaterStress	1.00	0.08	-0.08	0.04	0.05	-0.04	0.10	0.03
BankingAccess	0.08	1.00	0.66	0.80	0.49	0.81	0.81	0.65
RnDInvest	-0.08	0.66	1.00	0.72	0.46	0.59	0.49	0.43
GovtEffectiveness	0.04	0.80	0.72	1.00	0.49	0.69	0.62	0.71
ShippingIndex	0.05	0.49	0.46	0.49	1.00	0.49	0.46	0.09
FixedIntSubs	-0.04	0.81	0.59	0.69	0.49	1.00	0.83	0.46
InternetUsersPct	0.10	0.81	0.49	0.62	0.46	0.83	1.00	0.41
PoliticalStability	0.03	0.65	0.43	0.71	0.09	0.46	0.41	1.00
RuleOfLaw	0.05	0.78	0.70	0.94	0.39	0.65	0.57	0.78
RqdEduYears	-0.02	0.23	0.11	0.28	0.16	0.32	0.38	0.20
GovtEduSpendPctGDP	0.06	0.36	0.32	0.27	-0.02	0.12	0.17	0.33
MortalityFromDirtness	-0.12	NA	-0.23	-0.57	-0.27	-0.56	-0.76	-0.45
UHCServiceCoverage	0.10	0.76	0.54	0.70	0.46	0.73	0.82	0.51
HealthSpendPerCapita	0.01	0.66	0.71	0.72	0.43	0.75	0.64	0.46
GovtHealthSpendPerCapita	0.02	0.66	0.69	0.71	0.39	0.74	0.63	0.46
MigrantsPct	0.61	NA	0.21	0.43	0.17	0.35	0.37	0.36
FoodInsecurityPct	-0.08	-0.69	-0.49	-0.74	-0.47	-0.72	-0.77	-0.51
RuralPopulGrowth	-0.13	-0.44	-0.05	-0.31	-0.24	-0.12	-0.16	-0.31
VoiceAccountability	-0.14	0.65	0.54	0.75	0.13	0.54	0.44	0.68
Homicides	-0.09	-0.28	-0.31	-0.24	-0.23	-0.22	-0.20	-0.18

COMMENTS:

- Tables 6 through 9 show the correlation between all the features, from -1 (perfect negative correlation) to +1 (perfect positive correlation)
- The rules of thumb we are using are:
 - If absolute value of correlation is 0.9 or greater, one of those features will be dropped from statistical predictive models due to collinearity
 - If absolute value of correlation is between 0.7 and 0.9, we will evaluate collinearity impact (e.g. using VIF)
- This data is easier to deal with visually, which can be found on subsequent pages of this report

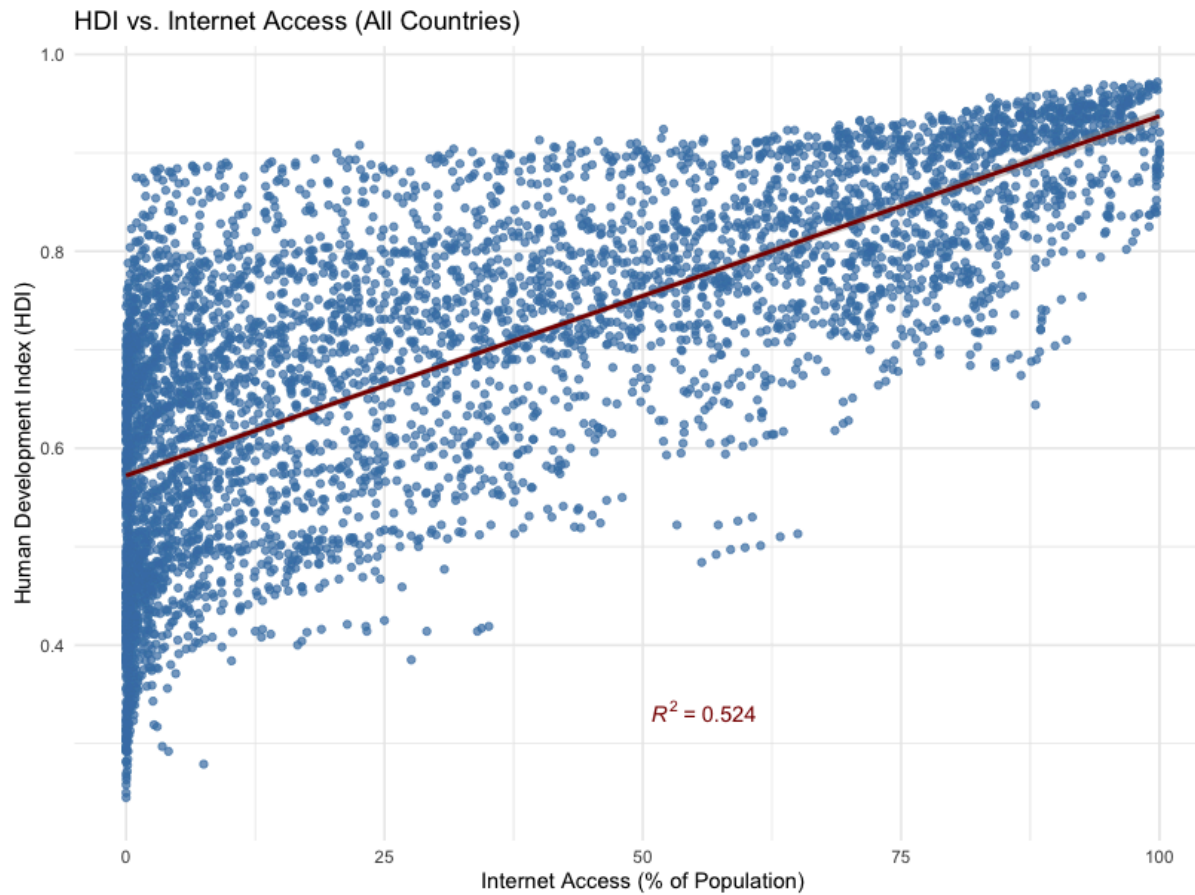
Table 8: Correlation Matrix: Columns 17 to 24

	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtness	UHCSserviceCoverage	HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct
HDI_Index	0.74	0.41	0.26	-0.80	0.92	0.64	0.63	0.41
IHDI_Index	0.77	0.39	0.28	-0.79	0.92	0.68	0.67	0.50
GiniCoeff	-0.68	-0.38	-0.28	0.80	-0.86	-0.57	-0.57	-0.41
FoodIndex	0.13	0.13	0.11	0.15	0.25	0.09	0.09	0.06
CorruptionScore	0.94	0.26	0.35	-0.42	0.61	0.73	0.72	0.43
ElectricAccess	0.50	0.42	0.20	-0.82	0.81	0.36	0.35	0.34
PopDensity	0.18	-0.01	-0.15	-0.07	0.13	0.26	0.29	0.37
PopInSlums	-0.63	-0.36	-0.33	1.00	-0.79	-0.48	-0.47	-0.38
WaterStress	0.05	-0.02	0.06	-0.12	0.10	0.01	0.02	0.61
BankingAccess	0.78	0.23	0.36	NA	0.76	0.66	0.66	NA
RnDInvest	0.70	0.11	0.32	-0.23	0.54	0.71	0.69	0.21
GovtEffectiveness	0.94	0.28	0.27	-0.57	0.70	0.72	0.71	0.43
ShippingIndex	0.39	0.16	-0.02	-0.27	0.46	0.43	0.39	0.17
FixedIntSubs	0.65	0.32	0.12	-0.56	0.73	0.75	0.74	0.35
InternetUsersPct	0.57	0.38	0.17	-0.76	0.82	0.64	0.63	0.37
PoliticalStability	0.78	0.20	0.33	-0.45	0.51	0.46	0.46	0.36
RuleOfLaw	1.00	0.26	0.29	-0.47	0.62	0.71	0.70	0.42
RqdEduYears	0.26	1.00	0.16	-0.35	0.45	0.28	0.26	0.11
GovtEduSpendPctGDP	0.29	0.16	1.00	-0.13	0.25	0.20	0.21	-0.01
MortalityFromDirtness	-0.47	-0.35	-0.13	1.00	-0.80	-0.35	-0.34	NA
UHCSserviceCoverage	0.62	0.45	0.25	-0.80	1.00	0.58	0.56	0.35
HealthSpendPerCapita	0.71	0.28	0.20	-0.35	0.58	1.00	0.97	0.38
GovtHealthSpendPerCapita	0.70	0.26	0.21	-0.34	0.56	0.97	1.00	0.37
MigrantsPct	0.42	0.11	-0.01	NA	0.35	0.38	0.37	1.00
FoodInsecurityPct	-0.68	-0.30	-0.22	0.70	-0.79	-0.52	-0.53	-0.39
RuralPopulGrowth	-0.31	-0.12	-0.08	0.49	-0.46	-0.10	-0.11	-0.05
VoiceAccountability	0.82	0.32	0.31	-0.34	0.50	0.57	0.56	0.17
Homicides	-0.31	0.09	-0.04	0.11	-0.07	-0.24	-0.26	-0.21

Table 9: Correlation Matrix: Columns 25 to 28

	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides
HDI_Index	-0.83	-0.24	0.59	-0.24
IHDI_Index	-0.85	-0.52	0.65	-0.30
GiniCoeff	0.84	0.50	-0.58	0.26
FoodIndex	0.14	-0.02	0.10	-0.01
CorruptionScore	-0.62	-0.30	0.77	-0.24
ElectricAccess	-0.78	-0.23	0.40	-0.06
PopDensity	-0.13	-0.03	0.08	-0.09
PopInSlums	0.77	0.48	-0.55	0.17
WaterStress	-0.08	-0.13	-0.14	-0.09
BankingAccess	-0.69	-0.44	0.65	-0.28
RnDInvest	-0.49	-0.05	0.54	-0.31
GovtEffectiveness	-0.74	-0.31	0.75	-0.24
ShippingIndex	-0.47	-0.24	0.13	-0.23
FixedIntSubs	-0.72	-0.12	0.54	-0.22
InternetUsersPct	-0.77	-0.16	0.44	-0.20
PoliticalStability	-0.51	-0.31	0.68	-0.18
RuleOfLaw	-0.68	-0.31	0.82	-0.31
RqdEduYears	-0.30	-0.12	0.32	0.09
GovtEduSpendPctGDP	-0.22	-0.08	0.31	-0.04
MortalityFromDirtness	0.70	0.49	-0.34	0.11
UHCSserviceCoverage	-0.79	-0.46	0.50	-0.07
HealthSpendPerCapita	-0.52	-0.10	0.57	-0.24
GovtHealthSpendPerCapita	-0.53	-0.11	0.56	-0.26
MigrantsPct	-0.39	-0.05	0.17	-0.21
FoodInsecurityPct	1.00	0.43	-0.54	0.39
RuralPopulGrowth	0.43	1.00	-0.31	0.01
VoiceAccountability	-0.54	-0.31	1.00	-0.04
Homicides	0.39	0.01	-0.04	1.00

Figure 1



COMMENTS:

- This scatterplot shows digital inclusion versus our wellbeing measure, with a trendline and R^2
- The positively sloped trendline shows that our basic hypothesis has promise as it does show correlation
- However, the low R^2 supports our prediction that predictability of wellbeing based on internet access alone is unlikely due to other factors

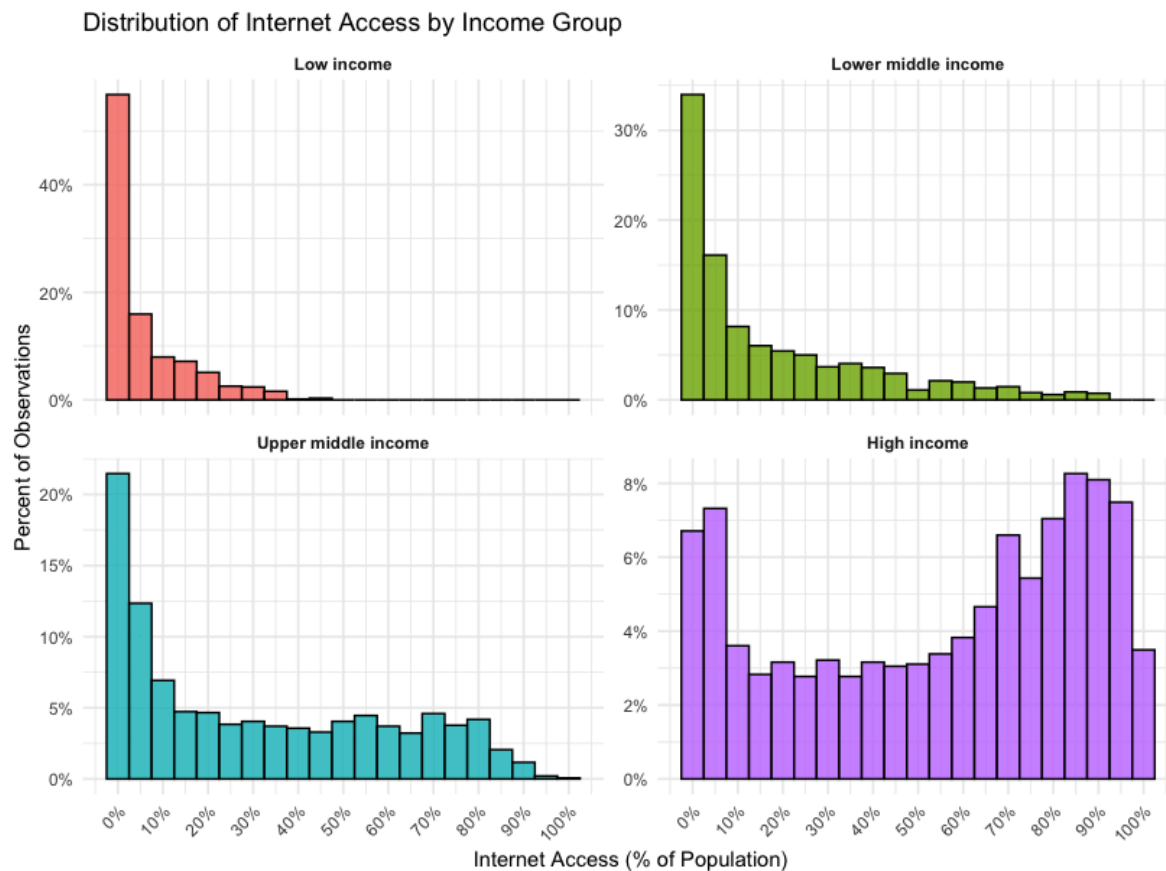
Figure 2



COMMENTS:

- This scatterplot shows the same relationship, but separated into income group cohorts
- We can observe the following:
 - The lower the income of the country cohort, the stronger the correlation between internet access and wellbeing (steeper trend line)
 - However, predictability is tied to income - the R² for lower income groups is much lower, suggesting (logically) that lower income countries have far more variables impacting wellbeing than higher income countries
- This plot supports the idea of clustering and / or controlling by income group - we'll see more on this later
- This plot and the previous both seem to show that much of the data (likely in earlier years when the internet was new) is clustered at around 0% internet access. We'll explore this further.

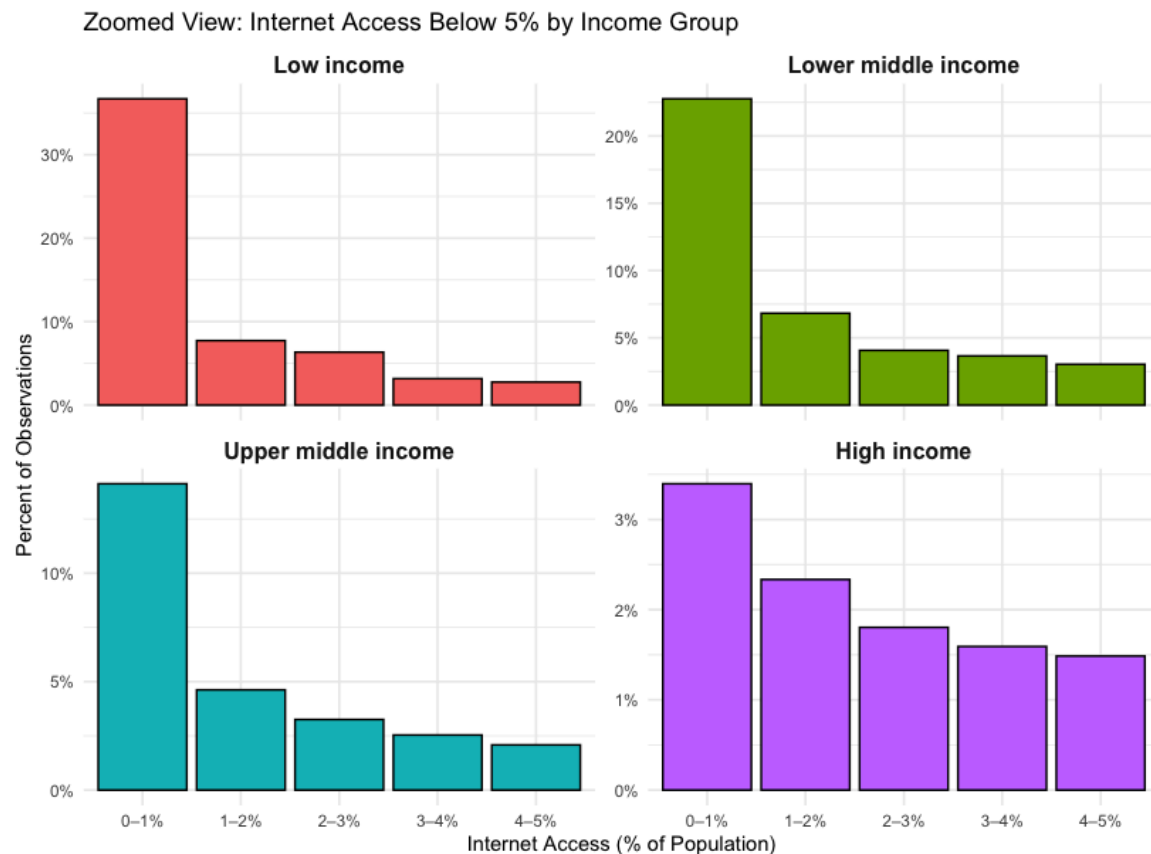
Figure 3



COMMENTS:

- These histograms show that we do indeed have a left skew issue, with data clustered near the 0% internet access side of the X axis
- However, what seems quite evident is that this problem is far more concentrated in low and middle income countries, which presumably were much later to the “internet party”
- From a modeling perspective, it’s useful to see how many of these data points are literally 0% since this left skew at 0% internet access may introduce considerable variance into our predictive models (which we see on the trendlines in the previous plots)

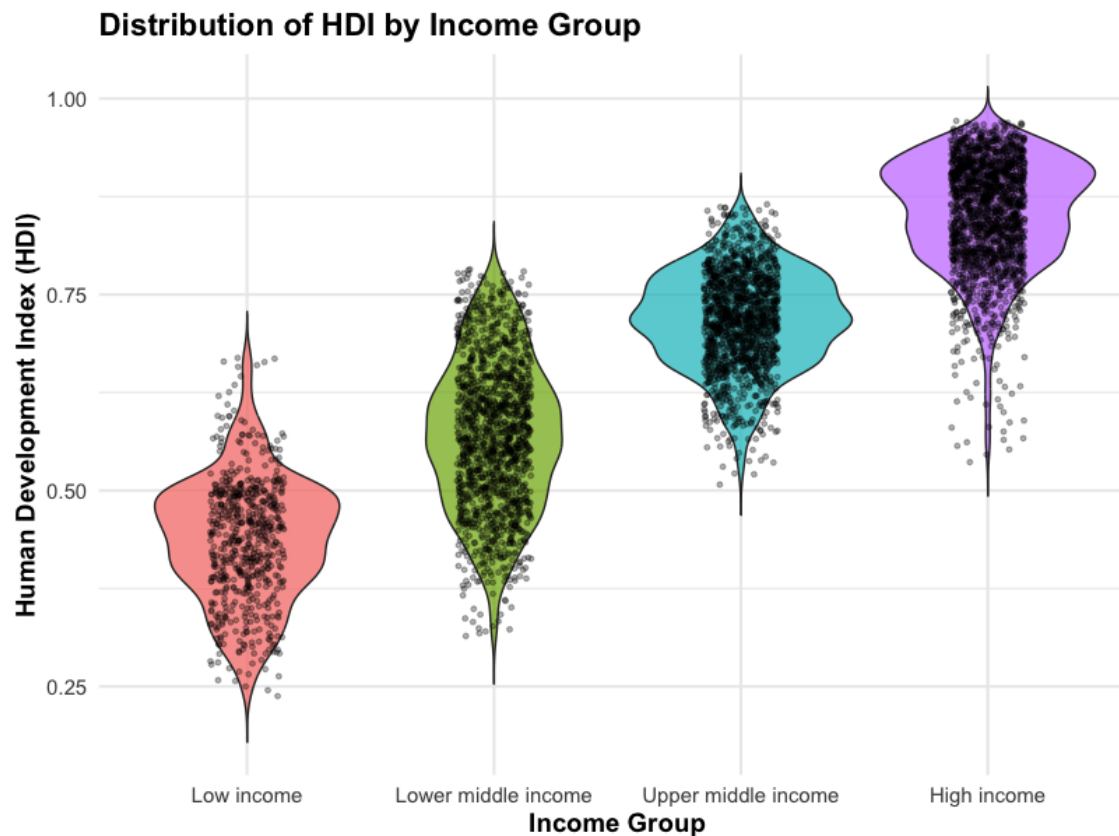
Figure 4



COMMENTS:

- Indeed, if we “explode” just the 0%-5% part of the plot, we do in fact observe that for all but the High Income countries, internet access of less than 1% (effectively 0) represents a very significant part of the data set
- This will need to be dealt with in our modeling approach to avoid massive variance caused by this data skew

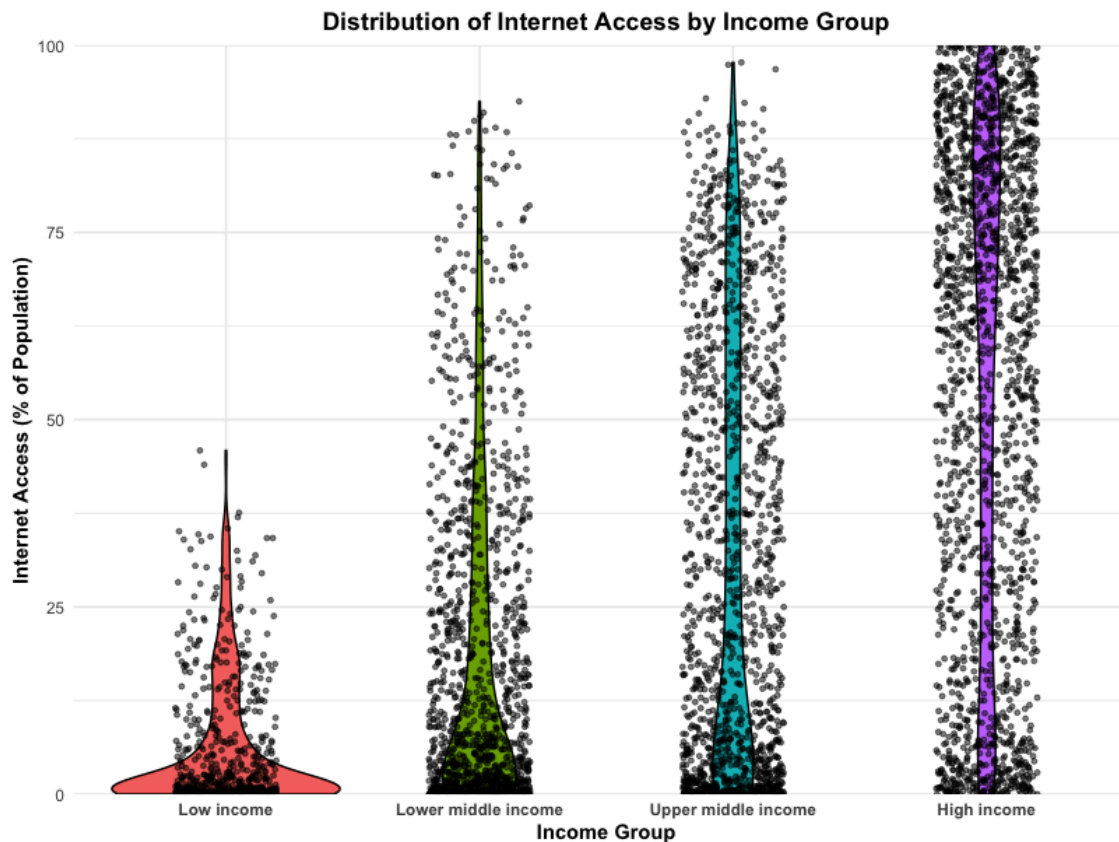
Figure 5



COMMENTS:

- This violin plot shows the extent to which wellbeing differs based on income (which is one of the three components of the HDI index we use to measure wellbeing)
- Not surprisingly, higher income groups show a higher overall level of wellbeing
- What we also see is that the distribution of HDI within each income group differs:
 - For the two lowest income categories, HDI can vary widely across its band relative to its median
 - The upper middle income group sees the variability in HDI decline, and still equally distributed around its median income
 - The High Income group's HDI band shows numerous outliers, skewed **NEGATIVELY** towards lower HDI which suggests that at higher overall national income levels, the impact of other factors like education and health can only worsen wellbeing, not improve it

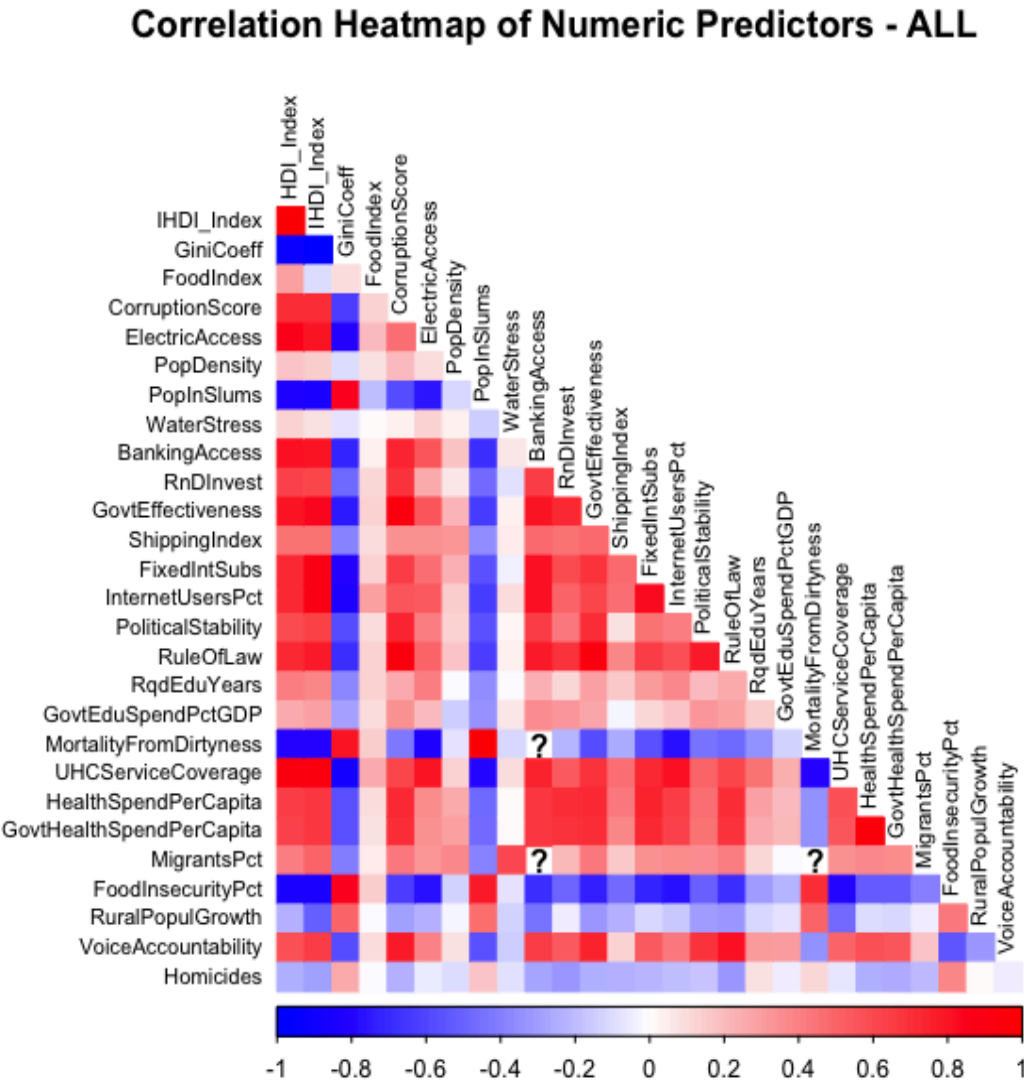
Figure 6



COMMENTS:

- This violin plot shows just how different internet access is based on income grouping
- Bearing in mind that the data analyzed contains time series, we can conclude:
 - Low income countries generally have shown slow if any progress at expanding access to the internet
 - Among wealthier nations, the violin plots get taller, thinner and more dispersed, representing the positive relationship between a country's income (and therefore its wellbeing) and its digital inclusion
 - HOWEVER- whether correlation = causation remains to be seen - could it simply be that citizens of wealthier countries simply are more likely to be able to afford internet access, and we have it backwards?

Figure 7

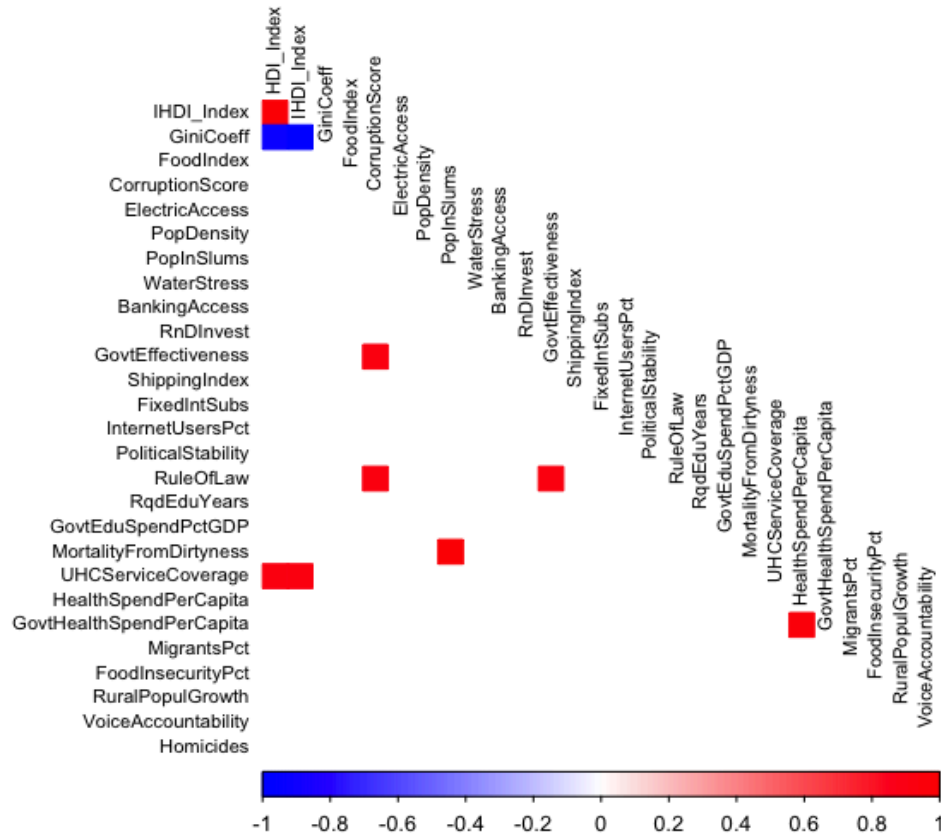


COMMENTS:

- This heatmap displays a comprehensive overview of the correlation coefficients for all numeric predictor variables in the dataset.
- A color gradient is used to represent the strength and direction of correlations. Dark blue indicates strong negative correlation, white indicates no correlation, and dark red indicates strong positive correlation.
- Several instances of missing correlation values are noted by question marks that suggest incomplete data for those specific variable pairs.
- This type of heatmap is valuable for understanding the interdependencies within a dataset and for identifying potential multicollinearity as well as features that might be good candidates for feature engineering.

Figure 8

Correlation Heatmap of Numeric Predictors - 90%+ correlation (DROP)

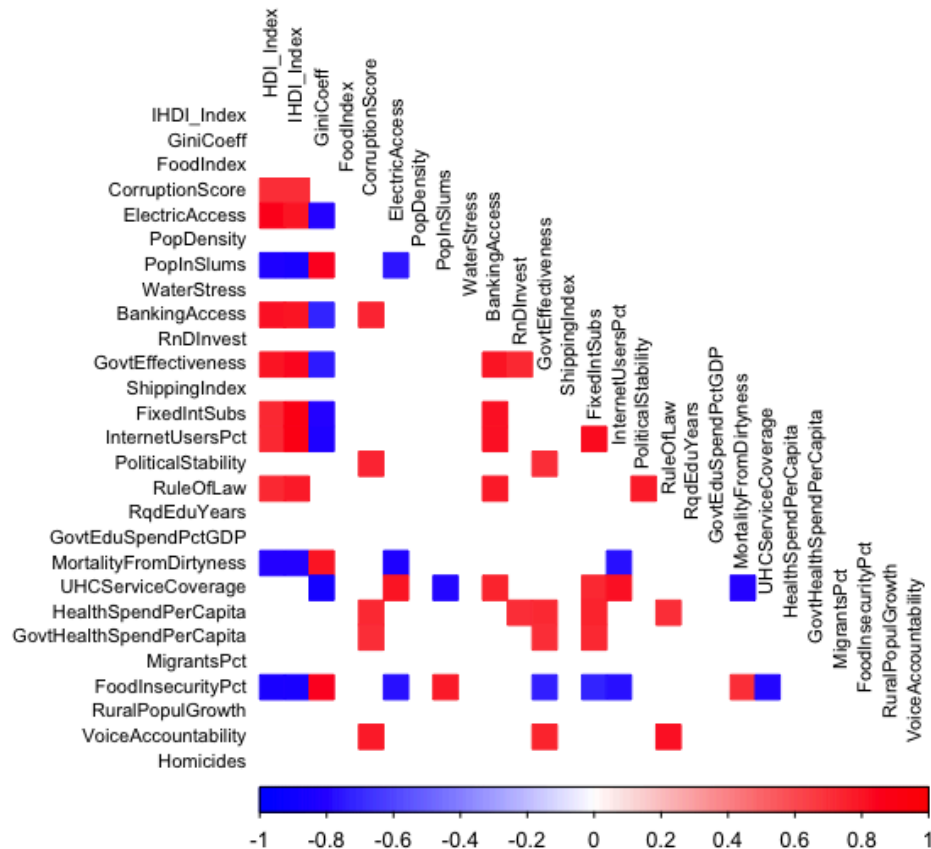


COMMENTS:

- This chart is the same correlation heatmap filtered to highlight numeric predictor variables with a correlation of 90% or higher.
- Correlations between features (NOT responses) of 90% or greater should lead to dropping one of the features in statistical models due to collinearity concerns
- Strong positive correlations (dark red) are evident, for example, between "IHDI_Index" and "HDI Index" as well as between both of these indexes and "UHCServiceCoverage"

Figure 9

Correlation Heatmap of Numeric Predictors - 70%-90%+ correlation (VIF)

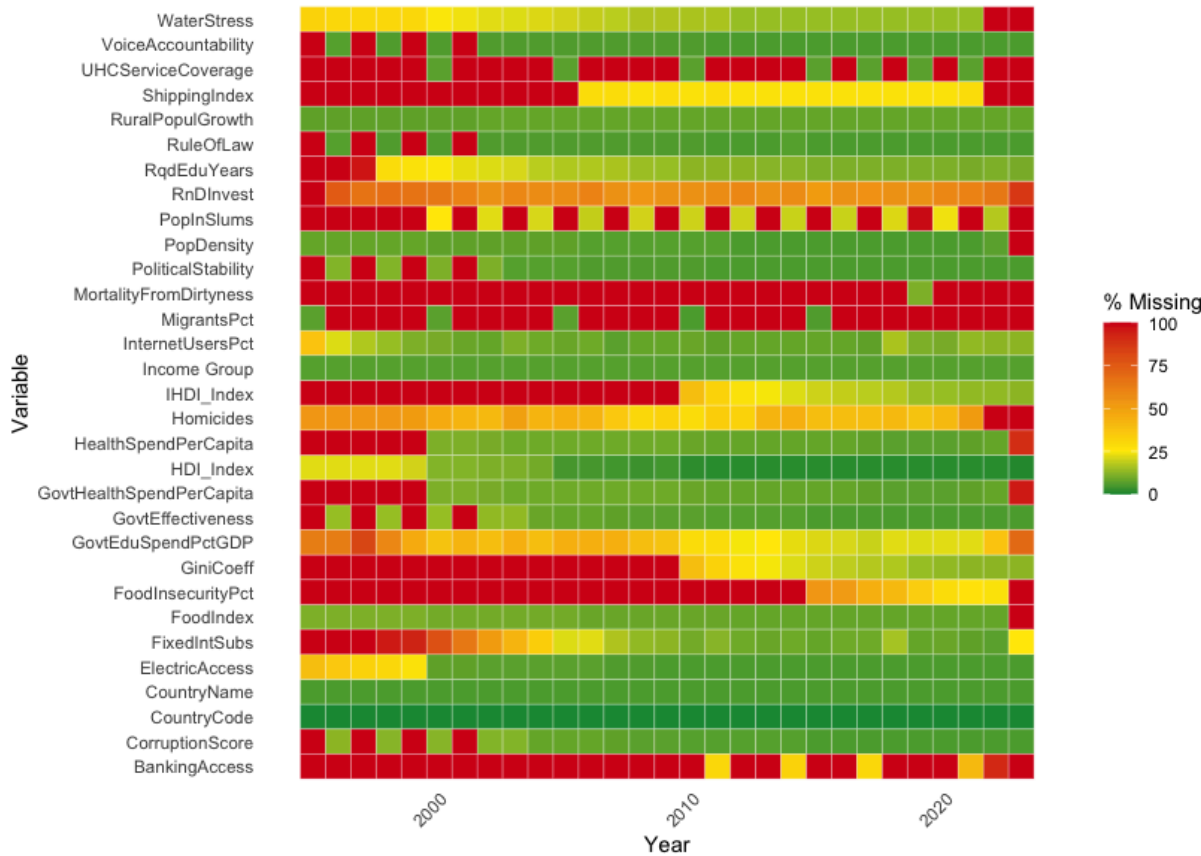


COMMENTS:

- This version filters variable pairs with correlations between 70% and 90%
- If used in statistical models, these features should be examined using VIF analysis to determine how much their collinearity may impact variance, and which (if any) should be dropped

Figure 10

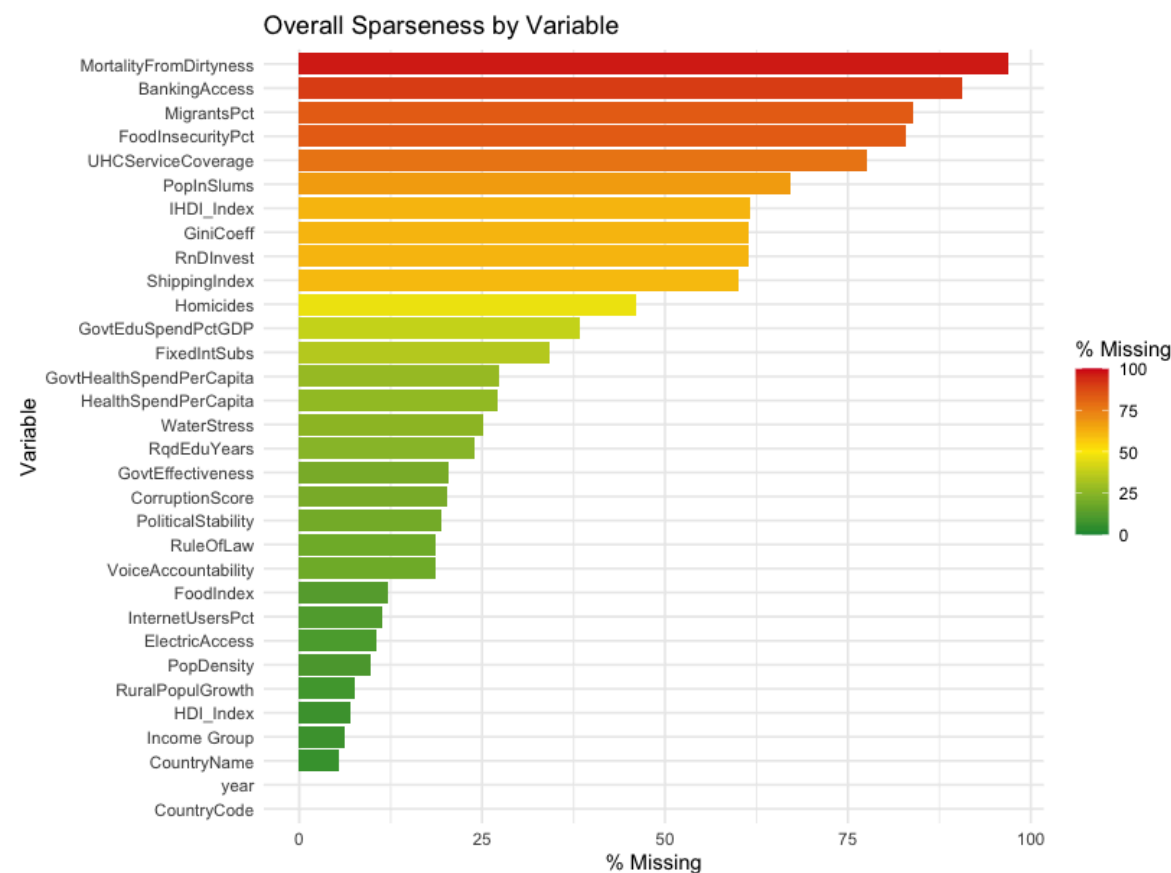
Missing Data Heatmap



COMMENTS:

- This heatmap chart exhibits the amount of missing data per feature (vertically) and over each year covered by our study (horizontally).
- There are several features that have a large amount of data missing over the complete range of years covered by this study such as “PopInSlums” and “MortalityFromDirtness”
- Generally the amount of data becomes more complete as the data collection period reaches more current dates.

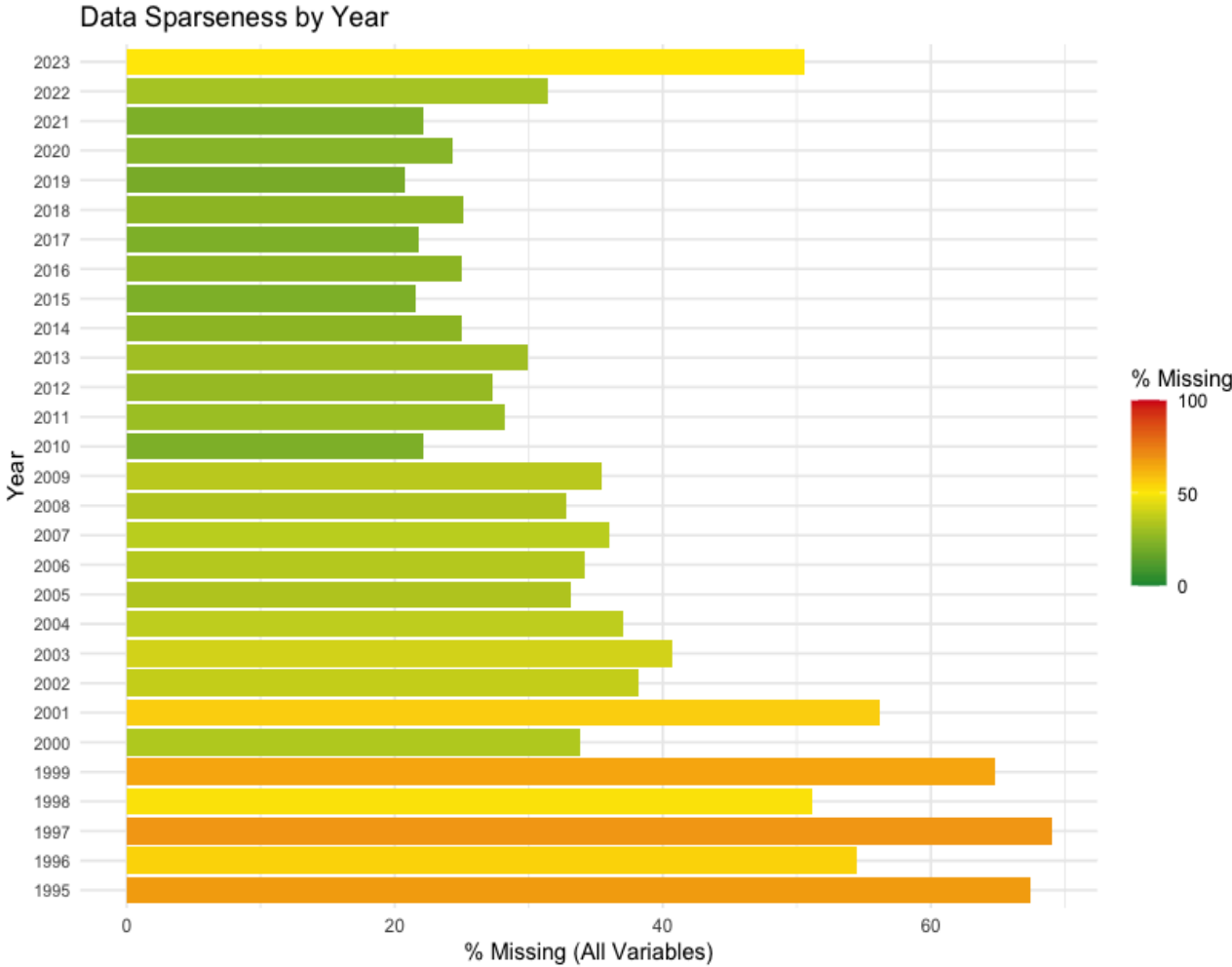
Figure 11



COMMENTS:

- This exhibit visualizes the sparseness of each variable as an indicator of which variables may warrant exclusion from the upcoming steps and which ones will need less imputation.
- This chart serves as an excellent overview of the nature of completeness of each variable, but it is important to take into consideration factors like the GiniCoefficient which is not collected each year that the data range accounts for.

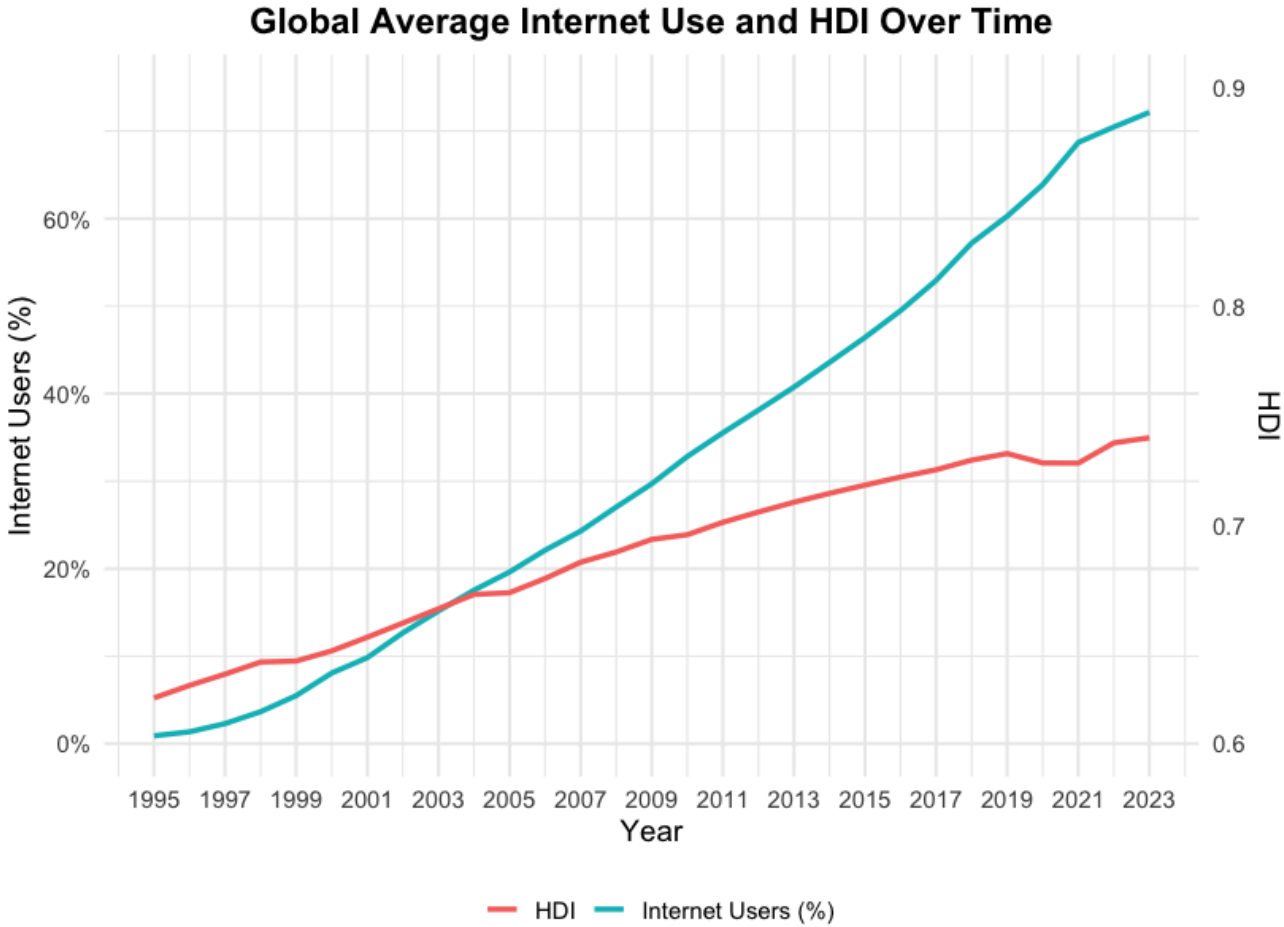
Figure 12



Comments:

- This chart visualizes the percentage of missing data across all variables for each year from 1995 to 2023.
- Earlier years exhibit the highest data sparseness, along with 2023 (the most recent year), with a significant portion of variables missing which is indicated by yellow-orange bars exceeding 50% missingness.
- Data completeness generally improves over time from 2004 onwards and shows considerably less missing data, due to broader collection of data globally by organizations.

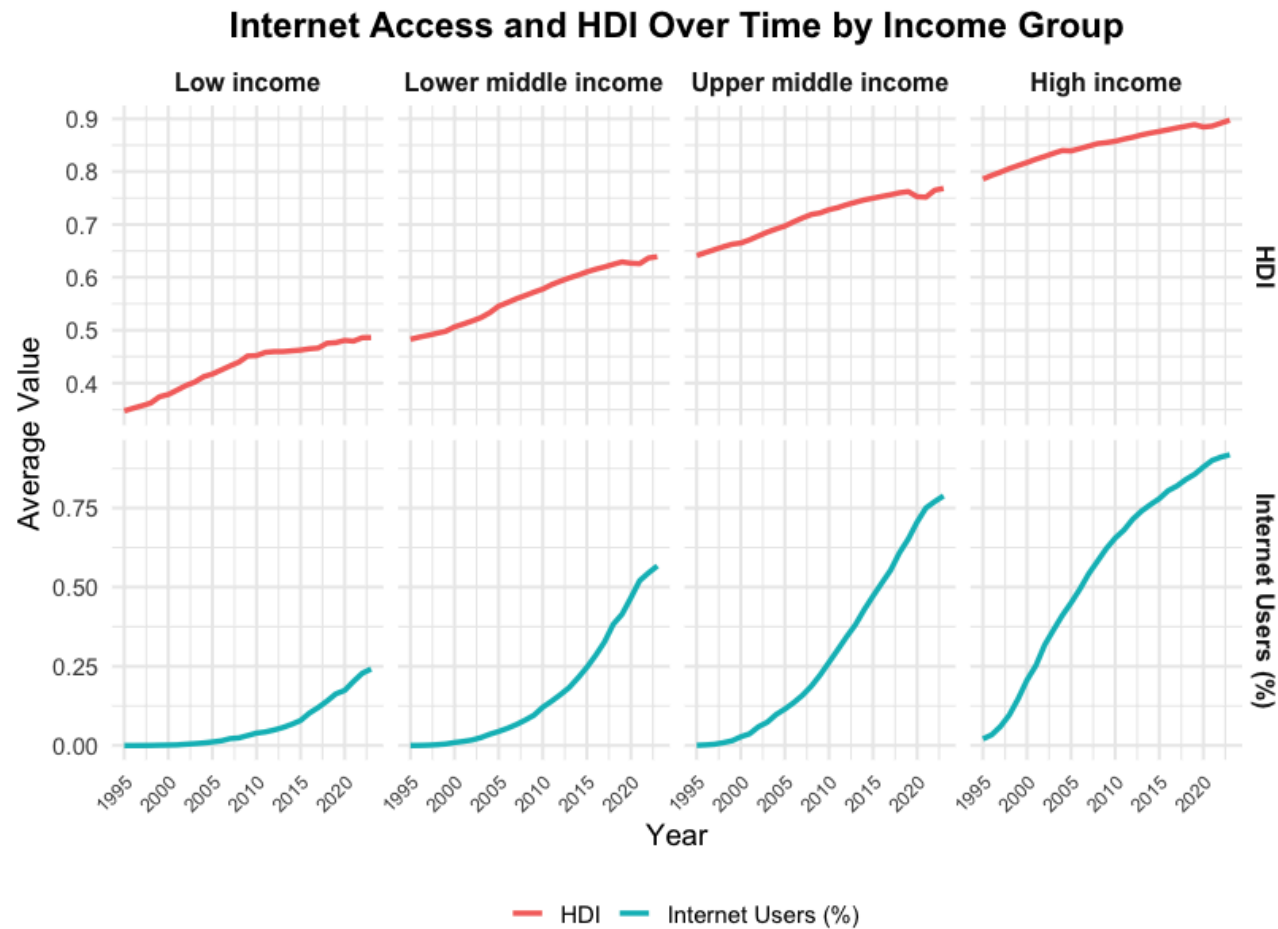
Figure 13



Comments:

- This chart visualizes the growth in average GLOBAL internet usage and HDI over time
- The relatively linear growth trajectories of both bode well for a predictive model correlating the two

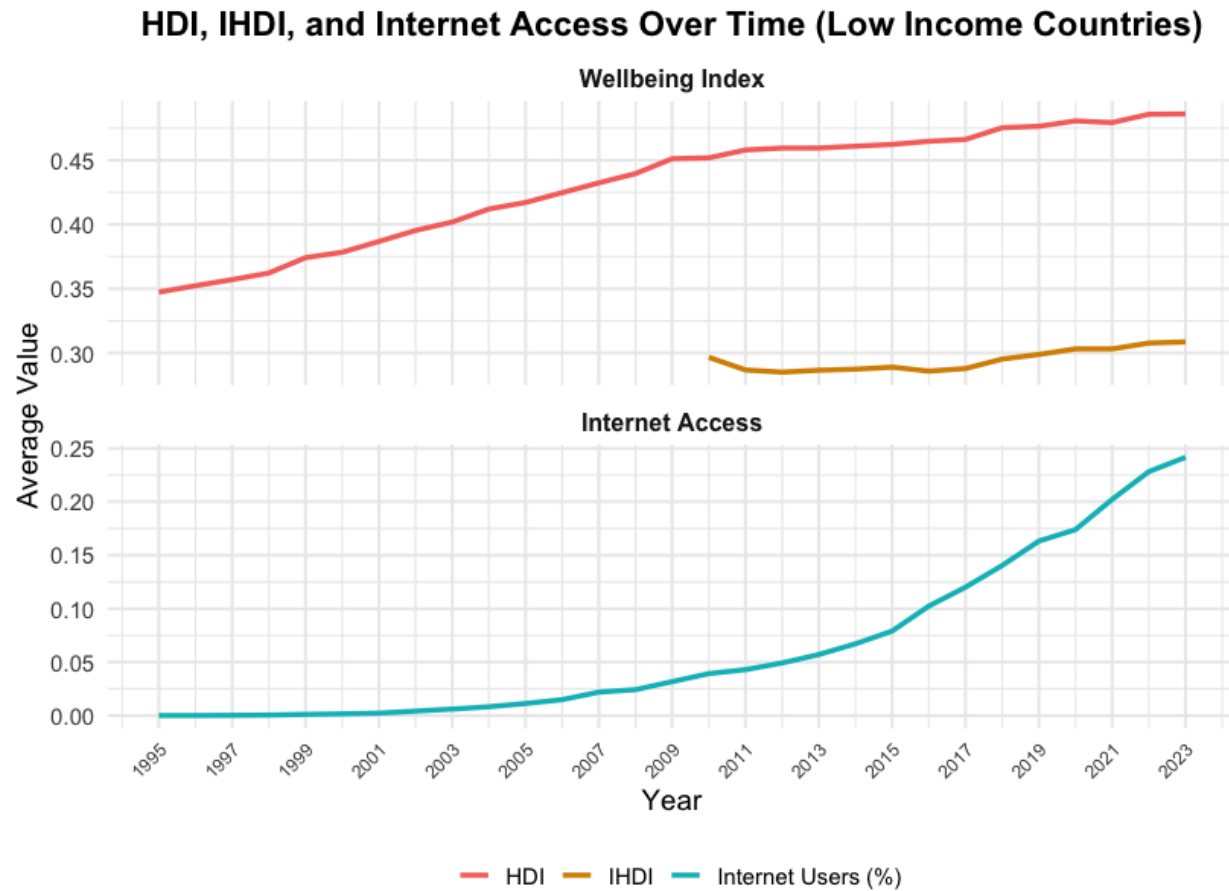
Figure 14



Comments:

- This chart breaks this data out by Income Group Cohort
- It's clear how much HDI is related to income given the differences in HDI across the income groups
- The higher the income cohort, the sooner internet access growth began and the steeper the adoption curve (good for our thesis)
- HOWEVER - there seems to be a flattening of the HDI growth trajectory for Low Income countries during the same period that internet usage growth finally starts to occur!

Figure 15



Comments:

- Indeed - that flattening is noticeable
- Could this actually REFUTE our thesis, suggesting that internet access for Low Income countries is actually a DRAG on wellbeing?
- Or could it be that either the global recession (which disproportionately impacted these countries) offset the impact of internet access....
- ...or that the growing impact of income inequality shown on this chart means the positive impact of internet access did not impact everyone equally?

Appendix A

Data Dictionary

Short Name	Category / "Topic"	Long Name	Type	Source
seriesCode ³	N/A	N/A	Predictor	ITU - Mobile
entityIso	N/A	N/A	Predictor	ITU - Mobile
dataValue	N/A	N/A	Predictor	ITU - Mobile
dataYear	N/A	N/A	Predictor	ITU - Mobile
seriesCode ⁴	N/A	N/A	Predictor	ITU - Fixed
entityIso	N/A	N/A	Predictor	ITU - Fixed
dataValue	N/A	N/A	Predictor	ITU - Fixed
dataYear	N/A	N/A	Predictor	ITU - Fixed
Country	N/A	N/A	Response	UNDP
Year ⁵	N/A	N/A	Response	UNDP
HDI Index	N/A	N/A	Response	UNDP
Gini Coefficient	N/A	N/A	Response	UNDP
Country Name	N/A	N/A	Identifier	World Bank (Data)
Country Code	N/A	N/A	Identifier	World Bank (Data)
Income Level	N/A	N/A	Predictor / Control	World Bank (Country)
Year ⁵	N/A	N/A	Identifier	World Bank (Data)
SE.SEC.CUAT.PO.ZS	Education: Outcomes	Educational attainment, at least completed post-secondary, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.SEC.CUAT.UP.ZS	Education: Outcomes	Educational attainment, at least completed upper secondary, population 25+, total (%) (cumulative)	Response	World Bank (Data)

³ Only keep series code i911mw which is users per 100 people, and rename to "Pop%"

⁴ Only keep series code i992b which is users per 100 people, and rename to "Pop%"

⁵ Year data may need to be created via a "pivot narrow" of the underlying data

Short Name	Category / "Topic"	Long Name	Type	Source
SE.TER.CUAT.BA.ZS	Education: Outcomes	Educational attainment, at least Bachelor's or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.DO.ZS	Education: Outcomes	Educational attainment, Doctoral or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.MS.ZS	Education: Outcomes	Educational attainment, at least Master's or equivalent, population 25+, total (%) (cumulative)	Response	World Bank (Data)
SE.TER.CUAT.ST.ZS	Education: Outcomes	Educational attainment, at least completed short-cycle tertiary, population 25+, total (%) (cumulative)	Response	World Bank (Data)
AG.PRD.FOOD.XD	Environment: Agricultural production	Food production index (2014-2016 = 100)	Predictor	World Bank (Data)
EN.POP.DNST	Environment: Density & urbanization	Population density (people per sq. km of land area)	Predictor	World Bank (Data)
EN.POP.SLUM.UR.ZS	Environment: Density & urbanization	Population living in slums (% of urban population)	Predictor	World Bank (Data)
SP.RUR.TOTL.ZG	Environment: Density & urbanization	Rural population growth (annual %)	Predictor	World Bank (Data)
EG.ELC.ACCS.ZS	Environment: Energy production & use	Access to electricity (% of population)	Predictor	World Bank (Data)
ER.H2O.FWST.ZS	Environment: Freshwater	Level of water stress: freshwater withdrawal as a proportion of available freshwater resources	Predictor	World Bank (Data)
FX.OWN.TOTL.ZS	Financial Sector: Access	Account ownership at a financial institution or with a mobile-money-service provider (% of population ages 15+)	Predictor	World Bank (Data)
SN.ITK.MSFI.ZS	Health	Prevalence of moderate or severe food insecurity in the population (%)	Predictor	World Bank (Data)
SH.XPD.CHEX.PC.CD	Health: Health systems	Current health expenditure per capita (current US\$)	Predictor	World Bank (Data)
SH.XPD.GHED.PC.CD	Health: Health systems	Domestic general government health expenditure per capita (current US\$)	Predictor	World Bank (Data)
SH.STA.WASH.P5	Health: Mortality	Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (per 100,000 population)	Predictor	World Bank (Data)
SP.DYN.LE00.IN	Health: Mortality	Life expectancy at birth, total (years)	Response	World Bank (Data)
SH.UHC.SRVS.CV.XD	Health: Universal	UHC service coverage index	Predictor	World Bank

Short Name	Category / "Topic"	Long Name	Type	Source
	Health Coverage			(Data)
IT.NET.BBND.P2	Infrastructure: Communications	Fixed broadband subscriptions (per 100 people)	Predictor	World Bank (Data)
IT.NET.USER.ZS	Infrastructure: Communications	Individuals using the Internet (% of population)	Predictor	World Bank (Data)
GB.XPD.RSDV.GD.ZS	Infrastructure: Technology	Research and development expenditure (% of GDP)	Predictor	World Bank (Data)
IS.SHP.GCNW.XQ	Infrastructure: Transportation	Liner shipping connectivity index (maximum value in 2004 = 100)	Predictor	World Bank (Data)
SI.POV.GINI	Poverty: Income distribution	Gini index	Response	World Bank (Data)
VC.IHR.PSRC.P5	Public Sector: Conflict & fragility	Intentional homicides (per 100,000 people)	Predictor	World Bank (Data)
CC.EST	Public Sector: Policy & institutions	Control of Corruption: Estimate	Predictor	World Bank (Data)
GE.EST	Public Sector: Policy & institutions	Government Effectiveness: Estimate	Predictor	World Bank (Data)
PV.EST	Public Sector: Policy & institutions	Political Stability and Absence of Violence/Terrorism: Estimate	Predictor	World Bank (Data)
RL.EST	Public Sector: Policy & institutions	Rule of Law: Estimate	Predictor	World Bank (Data)
VA.EST	Public Sector: Policy & institutions	Voice and Accountability: Estimate	Predictor	World Bank (Data)
SM.POP.TOTL.ZS	Social Protection & Labor: Migration	International migrant stock (% of population)	Predictor	World Bank (Data)

Appendix B

Use Of Generative AI For EDA

- As described above, we used Generative AI in two ways to assist with our EDA:
 - Ideation: Broad recommendations on what visualizations to produce, ESPECIALLY for sparseness and correlation
 - Coding resource: Ggplot code debugging and parameterization
- ChatGPT was selected based on research indicating that its training data makes it especially effective for EDA ideation and ggplot syntax support — particularly for layering and parameterizing complex plots.
- **Prompt Citation:**

We used ChatGPT to help ideate EDA methods and ggplot syntax and debugging. A shareable link to the full conversation is provided below, along with the exact prompt we submitted to initiate the exchange

View the conversation: <https://chatgpt.com/share/6888d4b4-ac50-800d-8f6a-0fb224fa7d1a>

I'm working on the EDA portion of our Capstone project. Our hypothesis is that population wellbeing (defined as either HDI or IHDI index as calculated by UNDP is correlated to how much of the population has internet access. We intend to do a 1:1 predictive modeling exercise looking at HDI vs internet access, but expect it to be noisy and imperfect from a predictive success perspective. We also intend to try to build a larger model to predict HDI using a number of data points from the World Bank's "development indicators" that we believe are likely to have great influence on population wellbeing. Within that model we hope to show that internet access is an important PART of a predictive model that predicts wellbeing.

The attached is a snippet of the data we have prepared. There are three identifier variables (country code, country name and year). There are two response variables (HDI and IHDI index).

The rest are potential predictors.

I have produced basic descriptive statistics in table form using both the summary and skim functions in R.

Now I need to produce additional EDA - ideally visually, that will help guide our feature selection, feature engineering and model building process.

I know that for statistical models, I can run a correlation matrix to determine collinearity (which is less important for machine learning models).

I also know that I should at minimum do a scatterplot (with a trendline through it) to see what the relationship between HDI and internet usage looks like, and I probably should also do that in a way that color-codes different groupings, for example, grouping by Income Level which is likely a significant control variable.

I also know that in the past I have used violin plots to show distributions by category / grouping, and histograms.

Can you give me some guidance based on this and the attached data set as to what would be really smart EDA plots and/or analyses that tie to our thesis? I'm particularly interested in whether there's a way to visually analyze collinearity, and what distributions, groupings and correlations I should be looking for - both the ones I may have mentioned and others.